

YORKTOWN USCG, VA, USA

WMO: 997315

Lat: 37.2265N

Lon: 76.4788W

Elev: 33

StdP: 14.68

Time Zone: -5.00 (NAE)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	21.1	25.1	N/A	N/A	N/A	N/A	N/A	N/A	32.3	37.4	28.0	36.7	11.5	340	0.500

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.4	91.1	N/A	88.8	N/A	86.8	N/A	N/A	N/A	N/A	N/A	N/A	N/A	8.7	240

Dehumidification DP/MCDB and HR

0.4%			1%			2%			0.4%		1%		2%		Extreme Max WB
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	(o)	(p)
27.8	23.9	20.2	DB	18.1	96.2	5.4	3.0	14.2	98.3	11.0	100.1	8.0	101.8	4.0	104.0	
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	60.7	40.6	42.2	48.3	59.7	68.1	76.1	79.7	78.5	72.9	63.4	52.1	46.2
	DBStd	15.64	9.97	9.10	8.92	8.11	6.83	5.12	3.81	3.53	4.85	7.15	7.48	8.66
	HDD50	979	325	252	145	12	0	0	0	0	0	3	63	179
	HDD65	3260	758	639	521	201	49	1	0	0	3	114	389	585
	CDD50	4900	33	34	93	303	560	782	921	883	686	418	126	61
	CDD65	1705	0	1	4	42	144	333	456	418	239	64	2	2
	CDH74	12773	0	1	25	268	780	2542	4267	3451	1214	221	2	2
	CDH80	3550	0	0	1	51	177	760	1396	916	223	26	0	0
Wind	WSAvg	9.9	10.2	10.7	11.6	11.2	10.1	9.0	8.5	8.4	9.4	10.2	10.0	9.5
Precipitation	PrecAvg	45.9	3.3	3.0	3.7	3.1	3.5	3.9	5.4	5.0	5.2	3.7	2.9	3.4
	PrecMax	62.6	5.7	9.4	9.5	5.6	5.1	9.0	8.9	11.6	17.6	8.5	10.5	7.0
	PrecMin	30.4	1.3	1.2	0.7	0.5	1.2	1.7	0.9	1.1	1.1	0.2	0.1	1.3
	PrecStd	8.2	1.3	1.7	1.7	1.3	1.0	1.8	2.1	2.8	3.7	2.0	2.2	1.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	69.8	73.0	77.7	85.3	88.3	93.9	95.7	93.0	89.1	83.1	73.9	73.3
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	65.9	67.4	72.1	80.6	84.9	89.6	91.4	89.5	85.6	78.9	69.9	69.3
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	61.6	63.6	67.3	76.9	81.4	87.0	89.0	87.2	82.4	76.2	66.4	65.1
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	10%	DB	56.8	57.2	63.1	72.8	78.3	84.5	86.8	85.0	80.1	73.8	63.5	60.4
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Mean Daily Temperature Range	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% DB	MDBR	12.7	13.3	13.5	14.1	12.3	11.6	11.4	10.2	9.6	11.0	12.1	12.3
		MCDBR	20.0	20.3	21.6	19.3	16.3	15.4	15.0	13.7	13.1	13.7	16.3	17.9
	5% WB	MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Clear-Sky Solar Irradiance	taub	0.306	0.317	0.345	0.392	0.434	0.485	0.515	0.511	0.410	0.353	0.322	0.309	
	taud	2.542	2.504	2.459	2.336	2.249	2.157	2.088	2.088	2.357	2.516	2.554	2.564	
	Ebn at Noon	282	292	292	282	271	256	247	245	267	275	274	273	
	Edh at Noon	24	29	33	39	43	48	51	50	36	28	24	23	
All-Sky Solar Radiation	RadAvg	738	977	1302	1652	1867	2015	1920	1733	1405	1111	835	655	
	RadStd	59	106	105	120	169	136	94	90	121	129	69	52	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (71 neighbors)	+0.83	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+290	+187

Nomenclature: See separate page